

LITTLE SWAN BALLET ACADEMY

Academy Management System

Database Design

Full Schema — All Tables & Relations

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Database Design — Little Swan Academy System

19 tables covering all system modules. All IDs are UUIDs. FK = Foreign Key. PK = Primary Key.

Section 1 — Core Tables

1. Students

One record per student, permanent. Never deleted.

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	Unique identifier
name_ar	Text		Full name in Arabic
name_en	Text		Full name in English
date_of_birth	Date		Age auto-calculated — never store age directly
parent_phone	Text		Primary contact number
level_id	UUID	FK	→ Levels table. Current ballet level.
status	Enum		active / frozen / inactive
notes	Text		Medical, behavioral, or administrative notes
enrollment_date	Date		Auto-set on record creation

2. Coaches

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
name_ar	Text		Arabic name
name_en	Text		English name
phone	Text		
email	Text		Used for login
password_hash	Text		Encrypted — never store plain text
hourly_rate	Decimal		EGP per hour. Set by Super Admin only.
can_login	Boolean		Allows Super Admin to toggle coach portal access

3. Staff / Users

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	

Field Name	Data Type	Key	Notes / Description
name	Text		
email	Text		Login email
password_hash	Text		Encrypted
role	Enum		super_admin / admin / staff

4. Halls

Single branch (Miami). Two halls only.

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
name	Text		e.g. Hall 1 / Hall 2 ةءاقلا 1 / ةءاقلا 2
capacity	Integer		Max students per session

5. Levels

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
name	Text		e.g. Pre-1, Pre-2, Grade-1, Grade-2...
order_num	Integer		Promotion sequence. Higher = more advanced.

Section 2 — Classes & Sessions

6. Classes (Fixed Weekly Schedule)

Template record. Sessions are generated from this.

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
name	Text		e.g. Grade-1 Saturday 2PM
level_id	UUID	FK	→ Levels
age_group	Text		e.g. 3-5 years, 6-9 years
default_coach_id	UUID	FK	→ Coaches. Can be overridden per session.
hall_id	UUID	FK	→ Halls. Hall 1 or Hall 2.
days_of_week	JSON		Array: e.g. [Saturday, Monday]
start_time	Time		e.g. 14:00

Field Name	Data Type	Key	Notes / Description
end_time	Time		e.g. 16:00
max_capacity	Integer		Overbooking prevention
is_active	Boolean		Soft-disable without deleting

7. Class_Students (Enrollment)

Many-to-many bridge. One student can be in multiple classes.

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
student_id	UUID	FK	→ Students
class_id	UUID	FK	→ Classes
enrolled_date	Date		
note	Text		Optional enrollment note

8. Sessions (Daily Occurrence)

Core daily unit. All attendance, payments, and coach records link to this.

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
class_id	UUID	FK	→ Classes. Parent class template.
date	Date		Actual date this session occurred
coach_id	UUID	FK	→ Coaches. Actual coach (may differ from class default).
staff_id	UUID	FK	→ Staff. Responsible employee on this day.
hall_id	UUID	FK	→ Halls. Inherited from class but stored for audit.
status	Enum		scheduled / completed / cancelled

Section 3 — Attendance & Subscriptions

9. Attendance

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
session_id	UUID	FK	→ Sessions
student_id	UUID	FK	→ Students

Field Name	Data Type	Key	Notes / Description
status	Enum		present / absent / make_up
original_session_id	UUID	FK	→ Sessions. Populated ONLY when status = make_up. Links to the original absent session.
recorded_by	UUID	FK	→ Staff
timestamp	DateTime		Auto-recorded on submission

10. Subscriptions

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
student_id	UUID	FK	→ Students
plan_id	UUID	FK	→ Subscription_Plans
total_sessions	Integer		e.g. 4
remaining_sessions	Integer		Decrement on each present / make_up attendance
start_date	Date		Date payment was made
end_date	Date		Calculated from plan or manually set
status	Enum		active / expired / suspended
payment_id	UUID	FK	→ Payments. The payment that created this subscription.

11. Freezes

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
student_id	UUID	FK	→ Students
start_date	Date		
end_date	Date		Optional — freeze can be open-ended
reason	Text		e.g. illness, travel, temporary departure
status	Enum		active / ended
created_by	UUID	FK	→ Staff

Section 4 — Transfers & Exams

12. Student_Transfers

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
student_id	UUID	FK	→ Students
from_class_id	UUID	FK	→ Classes
to_class_id	UUID	FK	→ Classes
status	Enum		pending / approved / rejected
requested_by	UUID	FK	→ Staff
approved_by	UUID	FK	→ Staff. Admin who made the decision.
request_date	Date		
decision_date	Date		
notes	Text		Reason or admin comment

13. Exams

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
name	Text		e.g. Mid-Year Exam / Final Exam
date	Date		
fee	Decimal		Auto-logged as payment on student enrollment
level_id	UUID	FK	→ Levels. Target level group for this exam.
created_by	UUID	FK	→ Staff

14. Student_Exams

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
student_id	UUID	FK	→ Students
exam_id	UUID	FK	→ Exams
result	Enum		pass / fail / pending
notes	Text		
recorded_by	UUID	FK	→ Staff
promoted_to_level_id	UUID	FK	→ Levels. Auto-populated on pass.

Section 5 — Finance

15. Payments

Central financial table. All revenue flows through here.

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
student_id	UUID	FK	→ Students
type	Enum		subscription / product / event / private / exam / enrollment
amount_due	Decimal		Total amount owed
amount_paid	Decimal		Amount collected
remaining_balance	Decimal		Calculated: amount_due – amount_paid
payment_method	Enum		cash / instapay
date	Date		
staff_id	UUID	FK	→ Staff. Who recorded this payment.
notes	Text		

16. Expenses

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
title	Text		e.g. Electricity, Internet, Cleaning Supplies
amount	Decimal		
date	Date		
staff_id	UUID	FK	→ Staff

Section 6 — Coach Attendance & Payroll

17. Coach_Attendance

Multiple records per session supported. Soft validation: invalid entries are flagged, not blocked.

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
session_id	UUID	FK	→ Sessions
coach_id	UUID	FK	→ Coaches
check_in_time	DateTime		Coach self-reported at session start
check_in_lat	Decimal		GPS latitude recorded at check-in

Field Name	Data Type	Key	Notes / Description
check_in_lng	Decimal		GPS longitude recorded at check-in
check_out_time	DateTime		Coach self-reported at session end
check_out_lat	Decimal		GPS latitude recorded at check-out
check_out_lng	Decimal		GPS longitude recorded at check-out
location_status	Enum		valid / invalid — based on distance from academy coordinates
hours_worked	Decimal		Auto-calculated: check_out_time minus check_in_time
staff_validated	Boolean		Staff cross-confirmation toggle
notes	Text		e.g. substitute, partial session, GPS flagged

17b. Staff_Attendance (Login / Logout Log)

Immutable audit log — records are never deleted. Every login event creates a new record.

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
staff_id	UUID	FK	→ Staff / Users
login_time	DateTime		Auto-recorded on system login
login_lat	Decimal		GPS latitude at login
login_lng	Decimal		GPS longitude at login
logout_time	DateTime		Auto-recorded on logout or session expiry
logout_lat	Decimal		GPS latitude at logout
logout_lng	Decimal		GPS longitude at logout
location_status	Enum		valid / invalid — based on academy radius check
hours_on_shift	Decimal		Auto-calculated: logout_time minus login_time
notes	Text		e.g. auto-logout due to session expiry

Section 7 — Inventory, Events & Private Sessions

18. Products (Leotards)

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
name	Text		Product name

Field Name	Data Type	Key	Notes / Description
type	Enum		ready_stock / made_to_order
size	Text		XS / S / M / L / XL / custom measurements
cost_price	Decimal		Purchase / tailor production cost
selling_price	Decimal		Price charged to student/parent
stock_qty	Integer		Current stock count (ready items only). Decrements on sale.

18b. Product_Orders (Made-to-Order)

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
student_id	UUID	FK	→ Students
product_id	UUID	FK	→ Products
size	Text		Custom measurements
status	Enum		pending / in_production / done / delivered
cost	Decimal		Tailor production cost
order_date	Date		
notes	Text		

19. Events

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
name	Text		e.g. End of Year Recital
type	Enum		recital / tv_show / workshop / competition
date	Date		
venue	Text		
price	Decimal		Enrollment fee per student
created_by	UUID	FK	→ Staff

19b. Event_Enrollments

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
event_id	UUID	FK	→ Events
student_id	UUID	FK	→ Students

Field Name	Data Type	Key	Notes / Description
payment_status	Enum		paid / partial / unpaid
payment_id	UUID	FK	→ Payments
notes	Text		

20. Private_Sessions

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
student_id	UUID	FK	→ Students
coach_id	UUID	FK	→ Coaches
date	Date		
duration_hours	Decimal		e.g. 1.0 / 1.5 / 2.0
fee	Decimal		Total amount charged
payment_id	UUID	FK	→ Payments. Auto-linked on creation.
notes	Text		

Section 8 — Settings Tables

21. Subscription_Plans

Field Name	Data Type	Key	Notes / Description
id	UUID	PK	
name	Text		e.g. Standard (4 sessions)
sessions_count	Integer		e.g. 4. Configurable by Super Admin.
price	Decimal		Default price — can vary based on level/group

22. System_Settings

Field Name	Data Type	Key	Notes / Description
key	Text	PK	Setting identifier e.g. max_freeze_days, absence_hide_threshold, academy_lat, academy_lng, allowed_radius_meters
value	Text		Setting value
description	Text		Human-readable explanation
updated_by	UUID	FK	→ Staff. Super Admin only.

Field Name	Data Type	Key	Notes / Description
updated_at	DateTime		

Relationship Summary

Tables	Type	Description
Students ↔ Classes	Many:Many	Via Class_Students. One student can be in multiple classes.
Classes → Sessions	One:Many	Each class generates many sessions (one per scheduled day).
Sessions → Attendance	One:Many	One attendance record per enrolled student per session.
Attendance (make_up) → Session	Self-Ref FK	make_up attendance references the original absent session via original_session_id.
Students → Subscriptions	One:Many	Multiple subscription cycles over time.
Sessions → Coach_Attendance	One:Many	Multiple coaches can check-in to one session.
Students → Payments	One:Many	All transaction types linked to the student.
Students → Freezes	One:Many	Multiple freeze periods possible over student lifetime.
Classes → Halls	Many:One	Each class assigned to Hall 1 or Hall 2. Conflict prevention enforced.
Payments → Subscriptions	One:One	Each subscription is created by one payment.
Payments → Event_Enrollments	One:One	Each event enrollment linked to one payment.
Payments → Private_Sessions	One:One	Each private session linked to one payment.
Sessions → Coach_Attendance (location)	One:Many	Each coach check-in stores GPS coords + location_status.
Staff → Staff_Attendance	One:Many	Every login/logout event creates an immutable record per staff member.